

Alysa Zhao

(469) 831-1833 | alyszhao111@tamu.edu | linkedin.com/in/alsaz | github.com/llysi | alysa.dev

EDUCATION

Texas A&M University

May 2028

B.S. Computer Science, Mathematics Minor

Honors: President's Endowed Scholar, National Merit Scholar

Relevant Coursework: Design & Analysis of Algorithms, Data Structures & Algorithms (C++), Computer Organization, Probability

EXPERIENCE

Infrastructure Engineering Intern

June 2026 – Present

Michaels Stores

Irving, TX

- Built a RAG pipeline that indexed millions of log events into a distributed Elasticsearch cluster, enabling rapid diagnosis of server failures through natural-language queries.
- Consolidated previously unused application, Windows, AIX, and network logs into one searchable index
- Automated disaster-recovery provisioning on IBM Cloud (PowerVS, VPC, Object Storage) with Terraform modules, Schematics, and CI/CD pipelines

AI Research Intern

Dec 2025 – Feb 2026

University of Texas at Dallas

Richardson, TX

- Contributed to a structural taxonomy of memory-augmented generation spanning four memory-organization categories by analyzing 20+ agentic memory systems
- **Publication:** D. Jiang et al., incl. **A. Zhao**. "Anatomy of Agentic Memory: Taxonomy and Empirical Analysis of Evaluation and System Limitations." arXiv:2602.19320, 2026

Peer Teacher, ENGR 102

Aug 2025 – Dec 2025

Texas A&M University

College Station, TX

- Guided 90+ students in Python fundamentals through office hours, code reviews, and exam prep; re-invited for future semesters.
- Developed Excel tracking tools to streamline grading and improve team coordination.

PROJECTS

Cartfish | TypeScript, Express, React, PostgreSQL

Oct – Dec 2025

- Full-stack grocery savings platform with a greedy set-cover optimizer for multi-store shopping lists and parallel price search
- Built a singleton OAuth 2.0 token manager with early-expiry refresh across 10 RESTful endpoints
- Designed a PostgreSQL schema with indexed foreign keys and a 7-day TTL cache over a 2,800+ store network

Launch Pad | Next.js, TypeScript — **Best Student Life Hack @ HowdyHack 2025**

Oct 2025

- Custom resume-to-job matching platform, converting resumes and scraped postings into structured JSON for LLM-based skill-gap analysis and roadmap generation
- Added ICS calendar export with configurable recurring study events for users to track learning progress

LEADERSHIP & ACTIVITIES

Web Officer

Sep 2025 – Present

Aggie Competitive Programming Club

- Competed in weekly algorithmic practice contests; built the front-end for the Spring programming contest site

Design Officer

Sep 2025 – Present

Texas A&M Computing Society (ACM Chapter)

- Collaborated on a reusable design system to cut marketing turnaround time; led visual identity for the chapter's hackathon

Royal Aces Dance Team

Aug 2024 – Present

President's Volunteer Service Award (3×, 500+ service hours)

TECHNICAL SKILLS

Languages: Python, C++, TypeScript/JavaScript, SQL, Java, Haskell

Infrastructure & Tools: Elasticsearch, Terraform, IBM Cloud, Docker, Git, Node.js, PostgreSQL, Supabase

ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, RAG pipelines, LLM agent orchestration, Matplotlib

Web: React, Next.js, Express, Tailwind CSS